

Module 1: Network + Network Fundamentals and building networks

Day 3 Switch and Router Configuration

Task 1:

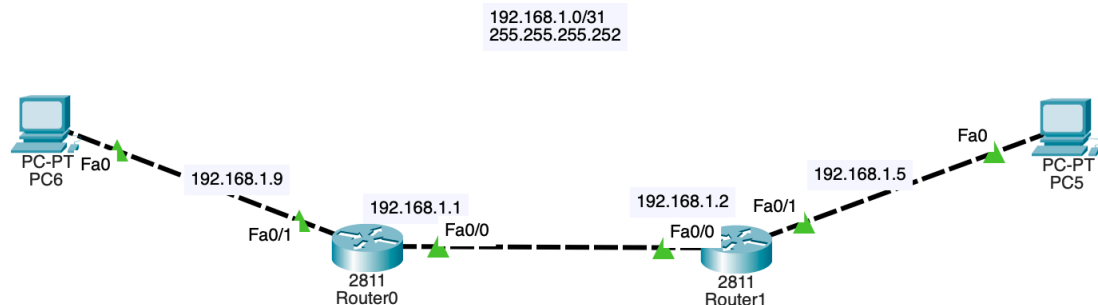
Creating VLANs 10 and 20 on the switch

```
Switch(config)#vlan 10
Switch(config-vlan)#name Sales
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Support
Switch(config-vlan)#exit
```

Task 2:

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Sales	active	Fa0/1, Fa0/2, Fa0/3
20	Support	active	Fa0/4, Fa0/5
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Task 3:



Task 4:

Sales VLAN pingging Support VLAN — why it fails

Why the ping fails: VLANs create separate broadcast domains at Layer 2 — even though both devices are physically connected to the same switch, the switch treats VLAN 10 and VLAN 20 as two entirely separate logical networks. A switch alone only forwards traffic within the same VLAN; it has no Layer 3 routing capability to move traffic between different VLANs, so the ping simply never leaves the Sales VLAN's broadcast domain.

How to fix it — two options:

Option A: Router-on-a-stick. Connect a single router interface to the switch via a trunk port, then create subinterfaces on the router — one for each VLAN — each with its own IP address acting as that VLAN's gateway.

Option B: Layer 3 switch. If your switch supports Layer 3 features (like a Catalyst 3560+), you can skip the external router entirely and create Switched Virtual Interfaces (SVIs) directly on the switch:

```
Switch(config)# ip routing
```

```
Switch(config)# interface vlan 10
```

```
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
```

```
Switch(config)# interface vlan 20
```

```
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
```

